The Register Is a Clock - AS

U06 · Practice · 24 written + 6 pen-first = 30 marks · Name: _____

Fresh authored questions. Work on paper. Use exactly eight bits unless another width is stated. Show working when requested. All marks are human marked; no computer is required.

1. Ours: name the representation first

(a) Represent -57 in eight-bit one's complement. Show your working. **[2]**

(b) Represent -57 in eight-bit two's complement. Show your working. **[2]**

2. Ours: one pattern, two readings

State the denary value of `1101 0101` as (i) one's complement and (ii) two's complement. One mark per value. **[2]**

3. Ours: the signed limits

State the smallest and largest integers that eight-bit two's complement can represent. **[2]**

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4. Ours: add a negative

Calculate 35 + (-62) using eight-bit two's complement. Show the operand encodings, addition working and eight-bit answer. **[2]**

5. Ours: subtract a positive

Calculate -27 - 19 using eight-bit two's complement. Show how subtraction becomes addition, your binary working and the eight-bit result. **[2]**

6. Ours: when the sign is a warning

(a) Add +91 and +52 using eight-bit two's complement. Show binary working and the eight stored result bits. **[2]**

(b) Explain why signed overflow occurs. Refer to the true sum and the signed range. **[2]**

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7. Ours: one byte, four contracts

Read `1001 0100` as (i) unsigned binary, (ii) two's complement, (iii) hexadecimal, and (iv) two BCD digits. One mark per value. **[4]**

8. Ours: from decimal digit to light

Describe a practical use of BCD in a decimal display, explaining how the digit encoding is used. **[2]**

9. Ours: a carry is not a verdict

```
  1111 1111
+ 0000 0001
-----------
1 0000 0000
```

Both operands use eight-bit two's complement. The carry-out is discarded. State whether signed overflow occurs and justify your decision using the signed values. **[2]**

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10. Ours: the signed reader - pen first

6 marks - design before code. Given `bits`, a valid eight-character string of 0s and 1s, calculate its two's-complement denary value. The first column has weight -128; the others have weights 64, 32, 16, 8, 4, 2, 1. Store the answer in the integer `total`. Use selection, a loop and arithmetic; do not use `int(bits, 2)`, `bin()` or formatting conversions.
Optional after writing: copy your Q10(b) statements into the supplied `binary_to_denary` wrapper in `challenge_U06_Converter.py`. Its header and return line are provided; writing a function is not assessed.

(a) Write numbered steps or draw a flowchart showing how you treat the first bit and accumulate the remaining column values. **[2]**

(b) Write Python statements that calculate total from bits. You do not need to write input, output, a function header or a return statement. **[3]**

(c) State the total your program should produce for `1110 0011`. **[1]**

Above and beyond - outside the 30 marks. Use four-bit two's complement. Encode -8, invert and add one, keeping only four bits. Explain why the result cannot be +8. Now write -8 in eight bits and negate it again. What did the extra room change?
Optional code extension: complete the supplied reverse-converter body too. The file checks each direction against known values; passing the checks is not a written-method mark.

Additional space - label the question number.